PHY · Static equilibrium and centre of mass — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Stability and centre of mass

Syllabus: S26-U3-SU-01 / S27-U3-SU-01

Frequency: 4 of 10 exam years; 7 marks in all.

Marks reward: Select the principle or relationship 2, Qualitative explanation 2, Diagram 2, Evaluate and conclude 1

Past questions: 2017R-6；2020C-21(c)；2021R-7；2022P-12(c)

Summary: This point asks why an object tips or stays upright, and it turns on where the centre of mass sits relative to its support base. Questions usually ask you to explain the restoring or tipping moment, find the position at which an object becomes unstable, or judge whether a stated arrangement will topple, often with a labelled diagram or a moments calculation.

Recognise: Look for words such as stable, unstable, tip over, topple, restoring moment, about to rotate, or stays upright. Any question that gives a support or pivot and a mass placed somewhere on a beam, plank or body, and asks what happens, is testing this point. A diagram of a rigid object resting on one or two supports is a strong signal.

Steps

1. Identify the pivot: for a body on a base, it is the edge of the base the object would rotate about if it tipped.
2. Locate the centre of mass and draw the weight force W = mg acting vertically through it. For a uniform object this is at the geometric centre.
3. Check whether the line of action of the weight falls inside the base. If it does, the net moment about the pivot restores the object; if it falls outside, the object tips.
4. For a calculation, take moments about the pivot: sum of clockwise moments = sum of anticlockwise moments, where moment = force x perpendicular distance.
5. Find the limiting case by setting the moments equal: the critical position of an added mass is where its moment just balances the moment of the rest of the object.
6. Write a conclusion stating the condition for stability in words and the numerical result with units.

Where the marks are

- A mark for choosing a sensible pivot, normally the edge of the support or base where the object would rotate.
- A mark for stating the moment condition clearly, such as the object is unstable when the moment that tips it exceeds the moment that holds it.
- A mark for a labelled diagram showing the weight's line of action and its position relative to the pivot.
- A mark for an explanation of the restoring effect: when the opposing force is removed, the weight's moment rights the object.
- A mark for a correct final evaluation that compares the two moments and states whether the object tips.

Pitfalls

- Using the centre of the whole object as the pivot instead of the edge of the base, which gives the wrong comparison.
- Forgetting that the weight of a body acts at its centre of mass, not at the point where a load is placed.
- Mixing up clockwise and anticlockwise moments, so the sign of the comparison is reversed.

- Stating only that the object will fall without showing the moment comparison or the line of action of the weight.
- Leaving out the units or failing to compare the two moment values numerically when asked to confirm a result.

Pattern 2: Torque about a pivot (τ = rF sinθ)

Syllabus: S26-U3-SU-02 / S27-U3-SU-02

Frequency: 8 of 10 exam years; 23 marks in all.

Marks reward: Select the principle or relationship 10, Substitution and working 8, Qualitative explanation 2, Diagram 2, Final answer 1

Past questions: 2016P-18(a); 2016P-18(b); 2017R-9; 2018R-4(b); 2019P-13(b); 2020C-21(dii); 2020P-18(a); 2020P-18(c); 2021R-7; 2022P-12(c); 2024P-16(a)

Summary: Torque questions ask for the turning effect of a force about a pivot, usually inside a static equilibrium or rigid-body balance problem such as ladders, drawbridges, scaffolds or supported beams. Most marks go to choosing a sensible pivot, writing a moment balance, and finding the effective lever arm from the given angle.

Recognise: An object is hinged, pivoted or resting on a support, with a force acting at a distance from the pivot and at an angle to the lever arm. Look for wording such as 'take moments about', 'balanced', 'just about to tip', 'in equilibrium', or questions asking why an object stays upright or how a turning effect changes when a load moves.

Steps

1. Choose a pivot that keeps unknown support or hinge forces out of the moment equation, usually a hinge or a support point.
2. Draw each force's line of action and find the lever arm r, the distance from the pivot to the point where the force acts.
3. Find the angle theta between the force F and the lever arm, then use tau = r F sin(theta). Equivalently, find the perpendicular distance r_perp = r sin(theta) and use tau = r_perp F.
4. Decide the turning direction of each moment and label it clockwise or anticlockwise, adopting one sign convention throughout.
5. For equilibrium, the total clockwise torque equals the total anticlockwise torque, so the net torque is zero. For a tipping condition, compare the torque from the weight with the torque from the support force.
6. Solve the resulting equation for the unknown and check that the sign and size of the answer are physically sensible.

Where the marks are

- A mark for choosing a sensible pivot and writing the moment balance (clockwise sum equals anticlockwise sum) before substituting numbers.
- Marks for finding the correct perpendicular distance or angle, using the sine or cosine of the given angle rather than the raw distance.
- A mark for the final numerical answer with correct units (N or N m) and sensible significant figures.
- Explanation marks for stating how the torque changes as the lever arm or force-to-lever angle changes, and linking that to a larger or smaller turning effect; stability answers need the restoring moment named explicitly.
- Diagram marks for labelling the line of action of the force and the perpendicular distance from the pivot.

Pitfalls

- Using the full distance from the pivot when the force is not perpendicular to the lever arm, forgetting the sin(theta) factor or using cos by mistake.

- Choosing a pivot that leaves two unknown forces in the moment equation, so the equation cannot be solved.
- Confusing the angle between the force and the lever arm with an angle to the horizontal or to a rope or beam.
- Forgetting the moment of a beam or plank's own weight acting at its centre of mass, or treating forces through the pivot inconsistently.
- Giving a final number without the moment equation, or writing an explanation without stating which way each torque turns.

Pattern 3: Equilibrium: ΣF = 0 and Στ = 0

Syllabus: S26-U3-SU-03 / S27-U3-SU-03

Frequency: 10 of 10 exam years; 126 marks in all.

Marks reward: Select the principle or relationship 48, Final answer 30, Substitution and working 28, Qualitative explanation 14, Diagram 5, Units, significant figures, direction 1

Past questions: 2016P-18(a); 2016P-18(b); 2016P-18(c); 2017R-11; 2017R-6; 2017R-9; 2018R-4(a); 2018R-4(b); 2019P-13(b); 2019R-6(a); 2019R-6(b); 2019R-7(a); 2019R-7(b); 2020C-21(dii); 2020P-13(a); 2020P-18(a); 2020P-18(b); 2020P-18(c); 2020R-5; 2021P-14(b); 2021P-14(ci); 2021P-14(cii); 2021R-6(a); 2021R-7...

Summary: This point tests static equilibrium of a rigid body: the net force and the net moment (torque) about any point must both be zero. It usually appears as a multi-step calculation on ladders, beams, scaffolds, drawbridges, seesaws and hinged supports, with some explain or diagram parts asking why a system is or is not balanced.

Recognise: Look for an object that is stationary or moving at constant velocity, a pivot, hinge or fulcrum, one or more supports or ropes, and a weight acting at a stated distance from a point. Phrases such as 'in equilibrium', 'just about to tip', 'balanced', 'remains at rest' or 'the tension in each rope' signal this point. A question asking for a force or tension at a hinge or support usually needs both the force balance and the moment balance.

Steps

1. Draw a clear free body diagram: mark every force (weight at the centre of mass, reactions at supports, tensions, hinge force) and where it acts.
2. Resolve any inclined force into horizontal and vertical components, for example Fx = F cos(theta) and Fy = F sin(theta), before writing equations.
3. Write the force balance: sum of horizontal forces = 0 and sum of vertical forces = 0.
4. Choose a convenient pivot (often a hinge or support where an unknown force acts, so that force drops out) and write the moment balance: sum of clockwise torques = sum of anticlockwise torques, using torque = r F sin(theta), where r is the distance from the pivot to the point of application.
5. Include the weight of the beam or plank itself, acting at its centre of mass, unless the question says it is negligible.
6. Solve the simultaneous equations, keeping track of which way each torque turns, and check the answer is physically sensible (positive tension, force pointing the right way).
7. For 'just about to tip' or 'unstable' questions, set the normal reaction at the support about to be lifted to zero, take moments about the other support, and solve for the critical position or load.

Where the marks are

- A mark for stating the equilibrium condition (sum of forces and sum of moments both zero) or choosing a sensible pivot.
- A mark for a correct moment equation with each distance and force identified, including the beam's own weight where relevant.
- A mark for resolving an inclined force into components or correctly using sin(theta) in the torque expression.
- A mark for correct simultaneous solution, followed by a final answer with units and a sensible

direction.

- For explain parts, a mark for linking the answer to the condition (for example, the restoring torque falls, so the force falls) and a separate mark for the consequence.

Pitfalls

- Using the wrong distance in a torque: the lever arm is the perpendicular distance, so the angle must be included when the force is not perpendicular to the beam.
- Forgetting the weight of the beam, plank or ladder itself when it is given as a mass.
- Mixing up which torques are clockwise and anticlockwise, so the moment equation gets a sign or side wrong.
- Taking moments about a point while leaving out a force that acts there, or ignoring the hinge force when writing the force balance.
- In explain parts, stating that the system is balanced without giving both the force reason and the moment reason.

Pattern 4: Static equilibrium contexts: ladders, beams, cables

Syllabus: S26-U3-SU-04 / S27-U3-SU-04

Frequency: 2 of 10 exam years; 7 marks in all.

Marks reward: Diagram 4, Final answer 2, Substitution and working 1

Past questions: 2016P-18(a)； 2019P-13(a)

Summary: This point examines static equilibrium of rigid bodies in familiar set-ups: a ladder leaning on a frictionless wall, a beam or seesaw pivoting about one point, bridges and cantilevers, and objects hung by cables. Questions usually ask for a fully labelled force diagram and then a moment or force calculation using the zero-net-force and zero-net-torque conditions.

Recognise: A diagram of a ladder, beam, seesaw, sign or bridge resting on a wall, pivoting on a support, or held by cables. Wording such as in equilibrium, at rest, does not slip or rotate, frictionless, or suspended by cables signals this point.

Steps

1. Draw the body on its own and mark every force: weight acting at the centre of mass, the normal reaction at a frictionless surface perpendicular to that surface, any pivot reaction, and cable tensions along each cable. Include friction only if the question gives it.
2. Label each force with a symbol and state its direction. A frictionless vertical wall gives a horizontal reaction only; a floor gives a vertical one.
3. Choose a pivot point, ideally where the most unknown forces act, so their moments about it are zero.
4. Apply the moment condition: sum of clockwise torques = sum of anticlockwise torques, where torque = F x d_perp and d_perp is the perpendicular distance from the pivot to the line of action of the force.
5. Apply the force conditions: sum of horizontal forces = 0 and sum of vertical forces = 0, to find any remaining unknowns.
6. Find perpendicular lever arms with trigonometry. For a ladder at an angle to the wall, use the ladder length and the angle that is actually given, and check whether it is measured from the wall or from the floor.
7. Check the result: the magnitude is positive, the direction matches your diagram, and the unit is newtons.

Where the marks are

- Diagram marks are awarded for each force drawn correctly, labelled and given its direction, including the weight at the centre of mass and the perpendicular reaction at a frictionless contact.
- Each separate force on the diagram earns its own mark, so none should be omitted, duplicated or drawn in the wrong direction.
- A mark is usually given for choosing a sensible pivot and writing the moment equation with the

correct perpendicular distances.

- A mark is often given for a trigonometric angle or lever-arm calculation that is needed before the moment equation can be set up.
- The final answer mark requires a numerical force with the correct unit and direction, consistent with the equation shown.

Pitfalls

- Using the full length of the ladder or beam as the lever arm instead of the perpendicular distance from the pivot to the line of action.
- Placing the weight at an end of the object rather than at the centre of mass, or leaving out the weight of the body altogether.
- Drawing a friction force at a frictionless wall, or forgetting that the wall reaction acts perpendicular to the wall.
- Confusing the angle to the wall with the angle to the floor, which swaps sine and cosine in the lever-arm calculation.
- Dropping units, or giving a force without stating its direction when the question asks for direction.